Object Oriented Programming (CSEG1044)

Unit 02 – Lecture 04: Inheritance, ‘super‘, Overriding, and Dynamic Dispatch

April 1, 2026

Sources: Schildt, *Java: The Complete Reference* (12e), McGraw-Hill (classes, inheritance, packages, interfaces).; Bloch, *Effective Java* (3e), Addison-Wesley, 2018 (design choices; interfaces vs abstract classes).; Oracle Java Tutorials: Classes and Objects + Interfaces + Packages (accessed April 1, 2026).; Java SE API docs (e.g., `Comparable`, `Runnable`) (accessed April 1, 2026).

Quick Links

Overview Key Topics Demo Summary

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Agenda

1 Overview

2 Key Topics

3 Demo / Example

4 Summary

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Lecture Snapshot

- **Unit focus:** Classes, Inheritance, Packages, and Interfaces

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Lecture Snapshot

- **Unit focus:** Classes, Inheritance, Packages, and Interfaces
- **Course thread:** Use Inheritance, ‘super‘, Overriding, and Dynamic Dispatch to strengthen the domain model, APIs, and access rules inside Campus Resource Manager.

Sources: Schildt, *Java: The Complete Reference* (12e), McGraw-Hill (classes, inheritance, packages, interfaces).; Bloch, *Effective Java* (3e), Addison-Wesley, 2018 (design choices; interfaces vs abstract classes).; Oracle Java Tutorials: Classes and Objects + Interfaces + Packages (accessed April 1, 2026).; Java SE API docs (e.g., `Comparable`, `Runnable`) (accessed April 1, 2026).

Recall Before You Start

- **Class design**: Inheritance helps only when the parent and child already have meaningful responsibilities. Main reference: `Unit 02 / Lecture 01 slides/notes`.

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Recall Before You Start

- **Class design**: Inheritance helps only when the parent and child already have meaningful responsibilities. Main reference: `Unit 02 / Lecture 01 slides/notes`.
- **Constructors**: The rule for super(...) depends on the constructor logic from the previous lecture. Main reference: `Unit 02 / Lecture 02 slides/notes`.

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Learning Outcomes

- Decide is-a (inheritance) vs has-a (composition).

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Learning Outcomes

- Decide is-a (inheritance) vs has-a (composition).
- Use `super(...)` and `super.method()` correctly.

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Learning Outcomes

- Decide is-a (inheritance) vs has-a (composition).
- Use `super(...)` and `super.method()` correctly.
- Override methods and explain dynamic dispatch with an example.

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Key Terms to Know

- **Inheritance**: a subclass extends a superclass (`extends`).
- **Superclass/Subclass**: parent/child in inheritance hierarchy.
- **Overriding**: subclass provides its own implementation of a superclass method (same signature).
- **Dynamic dispatch**: method call resolved at runtime based on the actual object type.
- **Upcasting**: treating a subclass object as a superclass reference, for example `Shapes=newCircle(...)`.
- **Composition**: has-a relationship (`Car` has an `Engine`).

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Study Flow for This Lecture

1. Refresh the prerequisite bridge before reading new ideas in isolation.

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1. Refresh the prerequisite bridge before reading new ideas in isolation.
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3. Trace the worked example and connect each line back to one concept frame.

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Study Flow for This Lecture

1. Refresh the prerequisite bridge before reading new ideas in isolation.
2. Study the main build in this order: *Inheritance and types (conceptual)* → *`super` usage* → *Overriding and dynamic dispatch*.
3. Trace the worked example and connect each line back to one concept frame.
4. Attempt the mini exercises before moving to the recap and exit check.

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Unit 02: Classes, Inheritance, Packages and Interfaces

- Lecture focus: Inheritance, ‘super‘, Overriding, and Dynamic Dispatch

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Today: Roadmap

- Inheritance and types (conceptual); "is-a" vs "has-a"

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- Inheritance and types (conceptual); "is-a" vs "has-a"
- ‘super‘ usage (constructors + member access)

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Today: Roadmap

- Inheritance and types (conceptual); "is-a" vs "has-a"
- ‘super‘ usage (constructors + member access)
- Overriding + runtime polymorphism (dynamic method dispatch)

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Is-a vs Has-a (design choice)

- Use inheritance for **is-a**: Dog is-a Animal.

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Is-a vs Has-a (design choice)

- Use inheritance for **is-a**: Dog is-a Animal.
- Use composition for **has-a**: Car has-a Engine.

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Is-a vs Has-a (design choice)

- Use inheritance for **is-a**: Dog is-a Animal.
- Use composition for **has-a**: Car has-a Engine.
- Avoid inheritance only for code reuse (creates strong coupling).

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super usage

- `super(...)` calls the parent constructor (must be first statement).

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super usage

- `super(...)` calls the parent constructor (must be first statement).
- `super.method()` calls parent implementation if overridden.

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Overriding + dynamic dispatch

- Overriding: same signature in subclass; use `@Override`.

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Overriding + dynamic dispatch

- Overriding: same signature in subclass; use `@Override`.
- Dynamic dispatch: method chosen at runtime based on actual object type.

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Overriding + dynamic dispatch

- Overriding: same signature in subclass; use `@Override`.
- Dynamic dispatch: method chosen at runtime based on actual object type.
- Example: `Shape s = new Circle(...);` then `s.area()` runs `Circle.area()`.

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Checkpoint and Reflection

Reflection prompt

Where would *Inheritance, 'super', Overriding, and Dynamic Dispatch* matter most in Campus Resource Manager, and which design choice would you justify using this lecture's ideas?

Quick self-check

Which part needs one more example before you move on: *Inheritance and types (conceptual)*, *`super` usage*, the worked example, or the recap?

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Example: Shape + Circle

```
class Shape {
    private final String color;
    Shape(String color) { this.color = color; }
    double area() { return 0.0; }
}

class Circle extends Shape {
    private final double r;
    Circle(String color, double r) {
        super(color);
        this.r = r;
    }
    @Override double area() { return Math.PI * r * r; }
}
```

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Dynamic dispatch demo

```
public class Demo {
    public static void main(String[] args) {
        Shape s = new Circle("red", 2.0);
        System.out.println(s.area());
    }
}
```

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Mini Exercises (2–3 mins)

1. Student–IDCard: is-a or has-a?

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1. Student–IDCard: is-a or has-a?
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Mini Exercises (2–3 mins)

1. Student–IDCard: is-a or has-a?
2. What must be first in a constructor: `super(...)` or other statements?
3. Overloading vs overriding: which is runtime?

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Watch-outs

- Choosing inheritance just to reduce code repetition quickly.

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Watch-outs

- Choosing inheritance just to reduce code repetition quickly.
- Forgetting that super(...) must run before subclass-specific initialization.

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Watch-outs

- Choosing inheritance just to reduce code repetition quickly.
- Forgetting that super(...) must run before subclass-specific initialization.
- Confusing overriding with overloading.

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Key Takeaways

- Use inheritance only for true is-a relationships.

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- Use inheritance only for true is-a relationships.
- `super(...)` initializes the parent part of an object.

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Key Takeaways

- Use inheritance only for true is-a relationships.
- `super(...)` initializes the parent part of an object.
- Overriding + dynamic dispatch enable runtime polymorphism.

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Checkpoint Questions

1. What is the difference between is-a and has-a?

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Checkpoint Questions

1. What is the difference between is-a and has-a?
2. Why do we use `@Override`?

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Checkpoint Questions

1. What is the difference between is-a and has-a?
2. Why do we use `@Override`?
3. What is dynamic dispatch?

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- Revisit the recall references if any older idea still feels weak.

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- Revisit the recall references if any older idea still feels weak.
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- Attempt the self-study practice and then answer the exit questions in full sentences.
- Apply the idea once inside Campus Resource Manager to make the concept stick.

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Next Lecture Preview

Next: Abstract classes, `final`, and interfaces (design choices)

- We will decide when to use abstract class vs interface.

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